

IHARQ BenchGuard Stretch C

Final Whole-Phase Closure Through Phase 02

Package: IHARQ_Final_Cumulative_Project_State_Through_P02_R1

Coverage: P00 -> P01 -> P02 / Layer 2

Closing run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Config SHA-256: 9e181d2e935d2e9674ca6e05572f49520ad0306a3761362b770f8bee8c78ce13

Scientific freeze: P02-PLANNED-SCIENTIFIC-EXECUTION-FREEZE-R5

Executive closure decision

The whole P02 lifecycle was checked as one chain: execution -> cumulative Protocol -> cumulative Phase Analysis -> Layer 0 -> cumulative Evidence Map -> cumulative Layer 10 -> downstream handoff. No mandatory scientific rerun was identified. Documentary and packaging integration was completed by creating this single cumulative consumer-facing handoff, explicit producer-consumer matrix, persistent-limitation handoff, canonical-artifact index, and external-retrieval contract.

P02_PHASE_AND_LAYER_CLOSURE_STATUS = GREEN_LIGHT_WITH_EXPLICIT

FINAL_CUMULATIVE_HANDOFF_ZIP_STATUS = VALIDATED (*set after archive verification; see final validation record*).

1. Closure basis

The accepted P02 runtime reports `bundle_contents_complete=true`, no partial failure, Stage 20 downstream readiness PASS, Stage 22 handoffs PASS with no manual reconstruction required, Stage 23 evidence sufficiency PASS with no missing items, and Stage 24 bundle PASS. The downstream-specific `Layer2ReadinessReport` has `compatibility_status=PASS`, no missing fields, and no blocking reasons.

The cumulative document lifecycle is also closed: source Protocol validation passes; the cumulative Phase Analysis final validation passes; the post-analysis Layer 0/Evidence Map/Layer 10 package passed 39/39 final checks.

2. Final cumulative authorities

- Protocol: IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1.
- Phase Analysis: cumulative P00-P02 scientific memory with deep P02 treatment.
- Layer 0: cumulative claim governance through P02; historical P00/P01 decisions preserved.
- Evidence Map: IHARQ-CUMULATIVE-EVIDENCE-MAP-THROUGH-P02-R1, current counts: 37 claims, 49 findings, 69 evidence nodes, 47 artifacts, 146 relationships.

- Layer 10: cumulative claim-safe, source-data-backed figures/tables and deterministic render specifications.

3. P02 execution and analytical parity

A0

Canonical accounting remains 678 terminal cells: 657 SUCCESS, 3 FAILED, 15 CONDITIONAL_SKIP, 3 DEPENDENCY_BLOCKED. The cumulative analysis does not reduce this to closure counts: it preserves dataset/model/budget/participant/statistical interpretation and the dataset-dependent decoder ranking. Layer 0 bounds the claims, the Evidence Map maps the evidence, and Layer 10 renders only governed views.

A4

Canonical accounting remains 1,218 terminal cells: 591 SUCCESS, 576 CONDITIONAL_SKIP, 9 INPUT_INCOMPATIBLE, 42 NOT_APPLICABLE_REPEAT_SLOT. The resource deviation remains explicit: the broader planned surface was reduced to the accepted single-refit-repeat/deep-anchor-budget surface. No downstream artifact is authorized to claim five-repeat stability, dense deep-budget response, or full Build Book replication equivalence.

Stage 11 / Stage 12

The correction history remains historical rather than retrospectively normalized. Stage 11 retains the mixed accepted primary lineage and the diagnostic R7 challenger. Stage 12 preserves the source/author-informed correction and the distinction between validation diagnostics and accepted scientific execution.

Stage 18S

Stage 18S remains post-hoc, descriptive, sensitivity-only evidence. It does not overwrite canonical Stage 18/G18 or create confirmatory multi-repeat evidence.

4. Downstream consumer readiness

P03 is the immediate governed next phase identified by the Build Book: **Calibration and Uncertainty**, involving L0/L1/L2/L3/L10 with A1/A2/A3. P02's consumer contract requires validated PredictionRecord partitions, ModelRegistryRecord lineage, score semantics, class order, A0 metrics, A4 identities, seed/config/budget/split lineage, Layer2ReadinessReport, and failures/limitations/DiagnosticOnly flags.

Observed P02 readiness substrate:

- PredictionRecord partition manifests: **1,014**.
- ModelRegistryRecord partition manifests: **672**.

- P03 compatibility status: **PASS**.
- Missing required fields: **0**.
- Blocking reasons: **0**.
- Stage 20/22/23/24: **PASS / PASS / PASS / PASS**.

Heavy rows/checkpoints are not duplicated into this handoff when the light project state intentionally externalized them. The handoff instead retains the exact local partition/index/schema evidence and immutable external retrieval contracts.

5. External artifact retrieval

Two symbolic HF credential classes remain required and separate: `IHARQ_HF_TOKEN_PRE_P02` and `IHARQ_HF_TOKEN_P02`. A single-token assumption is prohibited.

Preferred P02 continuation source:

- repository: `Csthv/p02-phase02-final-whole-working-20260814t052120z-b890ff92`
- immutable revision: `bc14961e14f2e48690e55df3577014275f9cbf30`
- workspace manifest SHA-256: `564bbb44e67d14da8457ad83d33651ded93dfc6b89ab90f10f78a36a20b8e8`
- per-object integrity index SHA-256: `540a0c3d4cbc686645ab1fccf088205c02c32a5c34e4b238c91f9c4ec`
- runtime checksum reconciliation: 68 local + 17021 externalized matched,
0 unmatched.

Archival fallback remains pinned separately at revision `257a407b5c7ea37b6c620863ee261c010c8f197c`. Live private remote availability is not claimed by this final closure; immutable local receipts/contracts control the handoff.

Security-preserving notebook handoff

The historical executed notebook contained three literal Hugging Face token strings. The raw notebook is therefore **not** included in this handoff ZIP. `execution/P02_Final_Executed_Notebook_SANITIZED.ipynb` is a transport-safe derivative in which only those literal credential strings are replaced by the governed symbolic `IHARQ_HF_TOKEN_P02`. `provenance/notebook_transport_redaction.json` preserves the original notebook SHA-256, redaction count, transport path, and the explicit statement that this is a security-only transport redaction with no scientific measurement change.

6. Persistent scientific limitations

These are nonblocking only because they are explicit and propagated:

- `PUBLIC_BENCHMARK_BINARY_MI_RESEARCH_ONLY` - Carry into downstream interpretation; do not silently broaden scope.
- `INHERITED_P01_DATA_SPLIT_PREPROCESS_WINDOW_SCOPE` - Carry into downstream interpretation; do not silently broaden scope.

- ONE_FROZEN_LOW_LABEL_SUBSET_PER_BUDGET - Do not infer repeated-subset sample-efficiency stability.
- NON_SUCCESS_TERMINALS_PRESERVED - Carry into downstream interpretation; do not silently broaden scope.
- POST_OBSERVATION_STAGE11_STAGE12_AMENDMENTS - Carry into downstream interpretation; do not silently broaden scope.
- STAGE11_MIXED_GRANDFATHERED_LINEAGE - Carry into downstream interpretation; do not silently broaden scope.
- A4_SINGLE_REFIT_REPEAT_AND_DEEP_ANCHOR_BUDGETS_ONLY - No five-repeat stability, dense deep-budget curve, or full-grid-equivalence claim.
- STAGE18S_POST_HOC_DESCRIPTIVE_ONLY - Keep Stage18S post-hoc/descriptive; never use as confirmatory replacement for Stage18.
- CONDITIONAL_SCORE_SEMANTICS - Carry into downstream interpretation; do not silently broaden scope.
- POST_RESTART_NOTEBOOK_RECONSTRUCTION - Carry into downstream interpretation; do not silently broaden scope.
- PRIVATE_HF_REMOTE_ACCESS_REQUIRES_AUTHORIZATION - Resolve symbolic credential class at runtime; never hardcode or assume one token.
- DUAL_SYMBOLIC_HF_CREDENTIAL_CLASSES - Resolve symbolic credential class at runtime; never hardcode or assume one token.
- RECORDED_RUNTIME_ENVIRONMENT_BOUND - Carry into downstream interpretation; do not silently broaden scope.
- DOWNSTREAM_CALIBRATION_POLICY_STRESS_EMBODIMENT_UNTESTED_BY_P02 - P03+ must test calibration/selective/uncertainty behavior; P02 does not establish it.

7. Required matrices

Master closure ledger summary

Status	Count
COMPLETE	16
COMPLETE_AFTER_CORRECTION	1
INHERITED_UNCHANGED	25

No ledger row is MISSING or BLOCKED. Full ledger: `handoff/master_closure_ledger.csv`.

Producer-consumer readiness

The complete handoff contains a **27-row producer-consumer matrix**, including **21 explicit Layer-2-produced consumer-facing product families**. The table below is a compact high-priority summary; the complete matrix is in `handoff/producer_consumer_matrix.yaml / .csv`, and the independent Layer

2 expected-product matrix is in `handoff/expected_L2_product_matrix.yaml`
/ `.csv`.

Record 1 - Artifact: P02-PREDICTION-SUBSTRATE - **Immediate consumer:** P03/L3 - **Availability:** POINTER_BACKED; partition manifests local; heavy rows recoverable from preferred workspace - **Integrity:** 1,014 partition manifests; Stage20 required fields PASS - **Ready:** YES

Record 2 - Artifact: P02-MODEL-REGISTRY - **Immediate consumer:** P03/L3 - **Availability:** LOCAL LIGHT RECORDS + external checkpoints if required - **Integrity:** 672 partition manifests; checkpoint hashes recorded - **Ready:** YES

Record 3 - Artifact: P02-A0 - **Immediate consumer:** P03 context - **Availability:** LOCAL - **Integrity:** analysis contract frozen - **Ready:** YES

Record 4 - Artifact: P02-A4 - **Immediate consumer:** P03 context - **Availability:** LOCAL - **Integrity:** Stage18/G18 PASS; governed counts - **Ready:** YES

Record 5 - Artifact: P02-L2-READINESS-P03 - **Immediate consumer:** P03 - **Availability:** LOCAL - **Integrity:** compatibility PASS; missing_fields=[]; blocking_reasons=[] - **Ready:** YES

Record 6 - Artifact: P02-FAILURE-INDEX - **Immediate consumer:** P03 - **Availability:** LOCAL - **Integrity:** parsed/hashed in final package - **Ready:** YES

Record 7 - Artifact: P02-EXTERNAL-PREFERRED - **Immediate consumer:** P03 - **Availability:** PRIVATE_EXTERNAL - **Integrity:** manifest and per-object index governed - **Ready:** YES

Record 8 - Artifact: P01-DATA-SUBSTRATE - **Immediate consumer:** P03 via P02 lineage - **Availability:** PRIVATE KAGGLE POINTERS - **Integrity:** Protocol pointers with manifest hashes - **Ready:** YES

Cross-document parity

Record 1 - Scientific item: Accepted run identity - **Protocol:** P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d - **Analysis / governance:** same / support provenance / same provenance / source registry - **Handoff:** P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d - **Consistency:** PASS

Record 2 - Scientific item: Config SHA-256 - **Protocol:** 9e181d2e...ce13 - **Analysis / governance:** same / support provenance / same / source registry - **Handoff:** 9e181d2e935d2e9674ca6e05572f49520ad0306a3761362b770f8bee8c78ce13 - **Consistency:** PASS

Record 3 - Scientific item: Scientific freeze - **Protocol:** P02-PLANNED-SCIENTIFIC-EXECUTION-FREEZE-R5 - **Analysis / governance:** same

/ support provenance / same / source registry - **Handoff:** P02-PLANNED-SCIENTIFIC-EXECUTION-FREEZE-R5 - **Consistency:** PASS

Record 4 - Scientific item: A0 closure - **Protocol:** 678 terminal; 657 SUCCESS / 3 FAILED / 15 CONDITIONAL_SKIP / 3 DEPENDENCY_BLOCKED - **Analysis / governance:** deeply analyzed / bounded decoder claims / A0 mapped / A0 figures/tables - **Handoff:** A0 inputs linked - **Consistency:** PASS

Record 5 - Scientific item: A4 closure - **Protocol:** 1218 terminal; 591 SUCCESS / 576 CONDITIONAL_SKIP / 9 INPUT_INCOMPATIBLE / 42 N/A repeat slots - **Analysis / governance:** deeply analyzed / resource-bounded claims / A4+deviation+Stage18S mapped / A4/limitation views - **Handoff:** A4 evidence linked - **Consistency:** PASS

Record 6 - Scientific item: Stage18S status - **Protocol:** post-hoc descriptive only - **Analysis / governance:** same / same boundary / separate supplement / visible limitation - **Handoff:** separate release handoff - **Consistency:** PASS

Record 7 - Scientific item: Dual credentials - **Protocol:** IHARQ_HF_TOKEN_PRE_P02 + IHARQ_HF_TOKEN_P02 - **Analysis / governance:** limitation/context / not claim-bearing / external provenance / not embedded - **Handoff:** explicit dual classes - **Consistency:** PASS

Record 8 - Scientific item: P03 readiness - **Protocol:** Layer2ReadinessReport in analysis contract - **Analysis / governance:** downstream-ready with limits / limitations propagated / readiness evidence / presentation only - **Handoff:** compatibility PASS; no missing fields - **Consistency:** PASS

P00 -> P02 harmony

Record 1 - Component: Authority/governance - **P00:** engineering foundation - **P01:** reused/extended - **P02:** reused/extended - **Current state:** single Governance V6.1 + cumulative authorities - **Consistency:** PASS

Record 2 - Component: Data identity - **P00:** schema/identity machinery - **P01:** 3 datasets + frozen split/preprocessing/windows - **P02:** consumes without mutation - **Current state:** P01 identities remain upstream data authority - **Consistency:** PASS

Record 3 - Component: Protocol - **P00:** preserved historical annex - **P01:** accepted execution contract - **P02:** accepted Layer2 execution + amendments - **Current state:** one cumulative Protocol through P02 - **Consistency:** PASS

Record 4 - Component: A4 - **P00:** conceptual/governance foundation - **P01:** matched 3.5s feasibility substrate; effectiveness deferred - **P02:** resource-constrained execution + Stage18S sensitivity - **Current state:** valid anchor evidence; generic improvement/stability not established - **Consistency:** PASS

Record 5 - Component: Claim governance - **P00:** historical dispositions

preserved - **P01:** historical dispositions preserved incl. A4 defer - **P02:** 10 P02 + cumulative claims governed - **Current state:** 37 current governed claims with historical lineage - **Consistency:** PASS

Record 6 - Component: Evidence Map - **P00:** predecessor preserved - **P01:** predecessor cumulative map preserved - **P02:** integrated current map - **Current state:** 49 unique findings / 37 claims / 69 evidence nodes - **Consistency:** PASS

Record 7 - Component: Layer10 - **P00:** prior artifacts preserved - **P01:** prior cumulative artifacts preserved - **P02:** P02 + cumulative views - **Current state:** claim-safe reproducible cumulative package - **Consistency:** PASS

8. Adversarial closure

- Missing P02 output: no unresolved required artifact found.
- Analysis loss: cumulative analysis retains the finalized detailed P02 body and its exact-number tables/machine-readable support.
- Evidence-map loss: orphan-claim and orphan-material-evidence reports are zero.
- Layer10 overclaim: Layer0/EvidenceMap/Layer10 parity remains closed; claim-safe captions and limitations are bundled.
- Earlier-phase corruption: P00/P01 IDs/history remain preserved rather than rewritten.
- Fresh-session P03 simulation: upstream schema/readiness/pointer information is available in this ZIP, with heavy bytes recoverable by pinned external contract.
- Future-paper simulation: Protocol, analysis, claims, map, figures, tables, exact source data, limitations, and provenance are all present.

9. Missing-artifact and blocker reports

UNRESOLVED_REQUIRED_ARTIFACTS = 0

DOWNSTREAM_BLOCKERS = 0

10. What is safe to do next

The finalized P00->P02 state is safe to consume as the upstream authority for the **P03 intake/planning step**, subject to the explicit limitations and external-access contracts bundled here. P03 should first validate/reuse the frozen P01/P02 inputs and Layer2ReadinessReport, then conduct its own governed Calibration and Uncertainty work (A1/A2/A3). This closure does not assert any P03 scientific result.

Final statuses

P02_PHASE_AND_LAYER_CLOSURE_STATUS = GREEN_LIGHT_WITH_EXPLICIT_NONBLOCKING_LIMITATIONS

FINAL_CUMULATIVE_HANDOFF_ZIP_STATUS = VALIDATED